

NICHOLAS SKIBA

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EDUCATION

Illinois Institute of Technology - Chicago, IL

August 2023 - December 2027 (Expected)

Co-Terminal Program: B.S. Mechanical Engineering / M.S. Mechanical and Aerospace Engineering

Minor: Engineering Graphics and CAD | B.S. GPA: 3.59 | M.S. GPA: 4.00

Coursework: Advanced CAD/CAM/CAE, Design of Machine Elements, Advanced Mechanics of Solids, Manufacturing Processes, Systems and Controls, Computational Mechanics, Dynamics, Thermodynamics, Fluid Mechanics, Materials Science

SKILLS

Design & Simulation: SolidWorks, Autodesk Inventor, AutoCAD, Mastercam, Revit, GD&T, ANSYS, FEA, Optimization, DFM

Programming & Development: Python, MATLAB, Arduino/C++, G-Code, Git/GitHub, VS Code, PlatformIO, Docker

Robotics & Controls: Embedded Firmware, Servo Control, Serial Communication, CAN Diagnostics, Mechatronic Systems

Testing & Instrumentation: Vector CANape, VNA, Oscilloscopes, RF Amplifiers, Signal Generators, Pressure Calibrators

Fabrication: 3D Printing, CNC/Machining, Laser Cutting, Welding, Plasma Cutting, Soldering, Mechanical Assembly

EXPERIENCE

Engine Controls Engineering Intern - Progress Rail, A Caterpillar Company - LaGrange, IL

May 2026 - August 2026

- Developed a repeatable pressure-sensor validation framework for engine-control applications, establishing test methodology, hardware configurations, measurement procedures, and structured data collection for six sensors
- Designed and assembled custom sensor wiring harnesses; verified continuity, grounding, connector pinouts, and electrical interfaces while integrating pressure calibration equipment and laboratory instrumentation
- Characterized pressure sensors for response, linearity, accuracy, repeatability, operating range, and sensor variation
- Built Python- and Excel-based workflows to analyze sensor data using automated regression, calibration modeling, and interpolation, evaluating linearity, residuals, repeatability, and sensor-to-sensor variation
- Performed system-level validation with engine control modules using Vector CANape and CAT ET to evaluate signal interpretation, diagnostics, boundary conditions, and fault-response behavior
- Documented test procedures, wiring configurations, analysis workflows, and engineering recommendations to support repeatable future sensor qualification and troubleshooting

Engineering Intern - Deep Coat Industries - Sugar Grove, IL

May 2025 - August 2025; December 2025 - January 2026

- Designed, built, and validated a 500 Hz–6.3 GHz EMI/RFI test system, including an anechoic chamber, cassette-style sample holder, transmitting/receiving antennas, and custom 3D-printed and machined fixtures for repeatable testing
- Automated instrument control, data acquisition, and engineering analysis using Python; developed workflows for dataset processing, visualization, regression, interpolation, and predictive shielding-performance analysis
- Developed theoretical shielding models (1 Hz–100 GHz) incorporating multilayer materials, recursive impedance calculations, frequency-dependent material behavior, and near-field/far-field electromagnetic formulations (Python)
- Led mechanical design of a compact 90-degree rotary mechanism, including full AutoCAD modeling, detailed part/assembly drawings, and manufacturing/assembly documentation
- Performed hands-on fabrication (plasma cutting, welding, soldering, etc.) and documented procedures for handoff

Business Owner/Operator - Triple Threat Services - IL

May 2024 - Present

- Founded and operate a mobile automotive detailing business, managing customer acquisition, scheduling, pricing, purchasing, financial tracking, customer communication, and service delivery, maintaining five-star client satisfaction

PROJECTS

Formula SAE Chassis Design - Illinois Tech

August 2026 - Present

- Developing a Formula SAE chassis/monocoque in SolidWorks, translating competition rules, Percy driver-template constraints, and structural requirements into CAD geometry, packaging, and preliminary structural analysis

Four-Bar EV Charging Arm (GitHub)

November 2025

- Developed automated EV charging arm using a four-bar linkage; optimized geometry in MATLAB (fmincon), reducing actuator torque 70%, with Arduino servo control and IR-triggered activation achieving <2 mm end-effector accuracy

Bio-Inspired Hexapod Robot (GitHub)

December 2025

- Designed and fabricated six-legged tripod-gait robot using Arduino and six servo actuators; integrated mechanical and electrical systems and validated coordinated walking over 10 m at 1.3 body lengths/s (0.24 m/s) through physical testing

Sustainable Foam-Core Chair (GitHub)

September 2025

- Devised, built, and tested a foam-core chair (about 462g) supporting 103 kg with optimized structure using Inventor FEA

ACTIVITIES & HONORS

Illinois Tech Men's Soccer

August 2023 - Present

- Student-athlete balancing engineering coursework with 20+ hours/week of training, competition, travel, and more

Dean's List - Illinois Institute of Technology

August 2023 - December 2025